

Understand every answer

Quadratics, functions, coordinate geometry and circles

YOUR AIM TODAY

Follow every solution from method choice to final answer, then repair your own first wrong step.

YOUR RULE

Open this only after completing the checkpoint. Compare one line at a time. If your first wrong line differs, stop copying and write the missing rule in your correction log.

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Start time: _____ Finish time: _____

Date: _____

Marking that produces learning

The final answer matters less than locating the first broken step.

Pass	Action
1. Compare	Put your work beside the worked solution.
2. Locate	Circle the first line where the two methods differ.
3. Classify	Label it: concept, formula, algebra, sign, notation or checking.
4. Repair	Write the correct rule without copying the full model.
5. Redo	Close this booklet and solve the entire part again.

MARK CODES

M means a method mark, A means an accuracy mark after a valid method, and B means an independent fact or statement.

Question	Topic	Marks
1	Core quadratics	8
2	Roots and discriminant	8
3	Inequalities and parameters	8
4	Functions	10
5	Coordinate geometry	12
6	Circles	14
Total		60

QUESTION 1

Core quadratic skills

Expand, complete the square and use the completed form.

1(a) Expand $(2x-3)(x+4)$.

CHOOSE THE METHOD

Multiply every term in the first bracket by every term in the second.

- $2x$ times x gives $2x^2$.
- $2x$ times 4 gives $8x$.
- -3 times x gives $-3x$.
- -3 times 4 gives -12 .
- Combine the middle terms: $8x-3x=5x$.

FINAL ANSWER - 2 MARKS

$2x^2+5x-12$. Award M1 for a correct expansion structure and A1 for simplification.

COMMON MISTAKE

Multiplying only the first terms and last terms. Four products are required.

1(b) Complete the square: $x^2-8x+11$.

CHOOSE THE METHOD

Half the x coefficient, square it, then compensate outside the bracket.

- Half of -8 is -4 , so begin with $(x-4)^2$.
- Expanding $(x-4)^2$ gives $x^2-8x+16$.
- We need $+11$, not $+16$, so subtract 5 .

FINAL ANSWER - 2 MARKS

$x^2-8x+11=(x-4)^2-5$. M1 for the correct bracket and A1 for -5 .

COMMON MISTAKE

Writing $(x-8)^2$. The number inside is half the x coefficient.

1(c) State the turning point.

CHOOSE THE METHOD

Read the vertex directly from $y=(x-a)^2+b$.

- The square is smallest when $x-4=0$, so $x=4$.
- At that point, the square is zero and $y=-5$.

FINAL ANSWER - 1 MARKS

Turning point $(4,-5)$. Award B1.

COMMON MISTAKE

Using $(4,5)$. The constant outside the square keeps its sign.

Exact roots

1(d) Solve $x^2-8x+11=0$ exactly.

CHOOSE THE METHOD

Use the completed-square form from part (b).

- Set $(x-4)^2-5=0$.
- Add 5: $(x-4)^2=5$.
- Square-root both sides: $x-4=\pm\sqrt{5}$.
- Add 4 to both sides.

FINAL ANSWER - 3 MARKS

$x=4+\sqrt{5}$ or $x=4-\sqrt{5}$. Award M1 for isolating the square and A1+A1 for both exact roots.

COMMON MISTAKE

Forgetting the negative square root. Squaring hides two possible values.

QUESTION 2

Roots and the discriminant

Factor when possible; use $D=b^2-4ac$ to describe roots.

2(a) Solve $3x^2-7x-6=0$.

CHOOSE THE METHOD

Factorise by finding numbers with product $3(-6)=-18$ and sum -7 .

- The numbers are -9 and $+2$.
- Split the middle term: $3x^2-9x+2x-6=0$.
- Group: $3x(x-3)+2(x-3)=0$.
- Factor: $(3x+2)(x-3)=0$.
- Set each bracket equal to zero.

FINAL ANSWER - 3 MARKS

$x=-2/3$ or $x=3$. Award M1 for a valid factorisation and A1 for each root.

COMMON MISTAKE

Turning $3x+2=0$ into $x=-2$. Divide by 3.

2(b) State and interpret the discriminant.

CHOOSE THE METHOD

Recall the expression inside the square root of the quadratic formula.

- For ax^2+bx+c , the discriminant is $D=b^2-4ac$.
- If $D=0$, $\sqrt{D}=0$, so the plus and minus formula gives the same answer.

FINAL ANSWER - 2 MARKS

$D=b^2-4ac$. $D=0$ means one repeated real root; the graph touches the x-axis. Award B1+B1.

COMMON MISTAKE

Saying there are no roots. No real roots occurs when $D<0$.

Repeated-root parameter

2(c) Find k for a repeated root in $x^2+6x+k=0$.

CHOOSE THE METHOD

A repeated root means set the discriminant equal to zero.

- Here $a=1$, $b=6$ and $c=k$.
- $D=6^2-4(1)(k)=36-4k$.
- Set $36-4k=0$.
- Therefore $4k=36$ and $k=9$.

FINAL ANSWER - 3 MARKS

$k=9$. Award M1 for $D=0$, A1 for substitution and A1 for solving.

COMMON MISTAKE

Using $b=6x$. The coefficient b is the number 6, not the term $6x$.

QUESTION 3

Inequalities and a parameter

Find boundary roots, then select regions using the parabola's sign.

3(a) Solve $x^2 - x - 12 < 0$.

CHOOSE THE METHOD

Factorise, mark the roots, and use the upward-opening shape.

- $x^2 - x - 12 = (x - 4)(x + 3)$.
- The boundary roots are $x = 4$ and $x = -3$.
- The coefficient of x^2 is positive, so the parabola is below the axis between its roots.
- The inequality is strict, so do not include the endpoints.

FINAL ANSWER - 3 MARKS

$-3 < x < 4$. Award M1 for roots, A1 for the correct region and A1 for strict endpoints.

COMMON MISTAKE

Choosing the outside regions. An upward-opening quadratic is negative between distinct roots.

3(b) Solve $2x^2 + 5x - 3 \geq 0$.

CHOOSE THE METHOD

Factorise, then choose where the upward-opening parabola is on or above the axis.

- $2x^2 + 5x - 3 = (2x - 1)(x + 3)$.
- The roots are $x = 1/2$ and $x = -3$.
- An upward-opening parabola is non-negative outside the roots.
- The symbol $\geq$ includes both roots.

FINAL ANSWER - 3 MARKS

$x \leq -3$ or $x \geq 1/2$. Award M1 for roots, A1 for regions and A1 for included endpoints.

COMMON MISTAKE

Writing $-3 \leq x \leq 1/2$. That is the region where the quadratic is non-positive.

No-real-root parameter

3(c) Find p if $x^2+px+9=0$ has no real roots.

CHOOSE THE METHOD

No real roots means the discriminant must be negative.

- Here $a=1$, $b=p$ and $c=9$.
- $D=p^2-4(1)(9)=p^2-36$.
- Require $p^2-36<0$, so $p^2<36$.
- Values whose square is below 36 lie between -6 and 6.

FINAL ANSWER - 2 MARKS

$-6<p<6$. Award M1 for $D<0$ and A1 for the interval.

COMMON MISTAKE

Writing $p<6$ only. Negative values below -6 also have squares greater than 36.

QUESTION 4

Functions, composites and inverses

Given $f(x)=2x-5$ and $g(x)=x^2+1$.

4(a) Find $f(-3)$.

CHOOSE THE METHOD

Substitute -3 in brackets wherever x appears.

- $f(-3)=2(-3)-5$.
- $2(-3)=-6$, then $-6-5=-11$.

FINAL ANSWER - 1 MARKS

$f(-3)=-11$. Award B1.

COMMON MISTAKE

Dropping the negative sign because the input was not placed in brackets.

4(b) Find $fg(x)$.

CHOOSE THE METHOD

$fg(x)=f(g(x))$; apply g first, then feed its output into f.

- $g(x)=x^2+1$.
- Replace the input of f by x^2+1 : $f(g(x))=2(x^2+1)-5$.
- Expand and simplify: $2x^2+2-5=2x^2-3$.

FINAL ANSWER - 2 MARKS

$fg(x)=2x^2-3$. Award M1 for correct order and A1 for simplification.

COMMON MISTAKE

Calculating $g(f(x))$. Composite order matters.

4(c) Find $gf(x)$.

CHOOSE THE METHOD

$gf(x)=g(f(x))$; apply f first, then square the entire result inside g.

- $f(x)=2x-5$.
- $g(f(x))=(2x-5)^2+1$.
- Expand: $(2x-5)^2=4x^2-20x+25$.
- Add 1: $4x^2-20x+26$.

FINAL ANSWER - 2 MARKS

$gf(x)=4x^2-20x+26$. Award M1 for substitution and A1 for expansion.

COMMON MISTAKE

Writing $4x^2+25$. The middle term -20x is essential.

Equations, inverses and one-to-one functions

4(d) Solve $fg(x)=15$.

CHOOSE THE METHOD

Use the expression found in part (b).

- $2x^2-3=15$.
- Add 3: $2x^2=18$.
- Divide by 2: $x^2=9$.
- Therefore $x=\pm 3$.

FINAL ANSWER - 2 MARKS

$x=-3$ or $x=3$. Award M1 for the equation and A1 for both roots.

COMMON MISTAKE

Giving only $x=3$. Both signs solve $x^2=9$.

4(e) Find f inverse.

CHOOSE THE METHOD

Undo the operations of f in reverse order.

- Write $y=2x-5$.
- Swap x and y : $x=2y-5$.
- Add 5: $x+5=2y$.
- Divide by 2: $y=(x+5)/2$.

FINAL ANSWER - 2 MARKS

$f^{-1}(x)=(x+5)/2$. Award M1 for rearrangement and A1.

COMMON MISTAKE

Writing $1/(2x-5)$. An inverse function is not a reciprocal.

4(f) Explain why g has no inverse on all real numbers.

CHOOSE THE METHOD

An inverse function requires the original function to be one-to-one.

- For $g(x)=x^2+1$, opposite inputs give the same output.
- For example, $g(1)=2$ and $g(-1)=2$.
- One output would therefore map back to two inputs.

FINAL ANSWER - 1 MARKS

g is not one-to-one on all real numbers. Award B1.

COMMON MISTAKE

Saying only 'because it is quadratic' without explaining the repeated outputs.

Coordinate geometry

A=(2,-1) and B=(8,7).

5(a) Find the gradient of AB.

CHOOSE THE METHOD

Use change in y divided by change in x in the same point order.

- $m = (7 - (-1)) / (8 - 2)$.
- The numerator is 8 and denominator is 6.
- Simplify 8/6 to 4/3.

FINAL ANSWER - 2 MARKS

Gradient 4/3. Award M1 for the formula and A1.

COMMON MISTAKE

Using 7-1 instead of 7-(-1). Subtracting a negative becomes addition.

5(b) Find the midpoint.

CHOOSE THE METHOD

Average the x-coordinates and y-coordinates separately.

- x midpoint = $(2+8)/2 = 5$.
- y midpoint = $(-1+7)/2 = 3$.

FINAL ANSWER - 1 MARKS

Midpoint (5,3). Award B1.

COMMON MISTAKE

Finding the change (6,8) instead of the midpoint.

5(c) Find the exact length AB.

CHOOSE THE METHOD

Use Pythagoras on the horizontal and vertical changes.

- Horizontal change = $8 - 2 = 6$.
- Vertical change = $7 - (-1) = 8$.
- $AB = \sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100}$.

FINAL ANSWER - 2 MARKS

AB=10. Award M1 for the distance calculation and A1.

COMMON MISTAKE

Adding 6+8. Distance is the hypotenuse, not the sum of perpendicular changes.

Line and perpendicular bisector

5(d) Find equation AB.

CHOOSE THE METHOD

Use $y - y_1 = m(x - x_1)$ with $m = 4/3$ and either point.

- Using $A = (2, -1)$: $y - (-1) = (4/3)(x - 2)$.
- So $y + 1 = (4/3)x - 8/3$.
- Subtract 1: $y = (4/3)x - 11/3$.

FINAL ANSWER - 3 MARKS

$y = (4/3)x - 11/3$. Equivalent: $4x - 3y - 11 = 0$. Award M1+A1+A1.

COMMON MISTAKE

Substituting $x = 2$, $y = +1$. Point A has $y = -1$.

5(e) Find the perpendicular bisector.

CHOOSE THE METHOD

A perpendicular bisector passes through the midpoint and has negative reciprocal gradient.

- AB has gradient $4/3$, so perpendicular gradient is $-3/4$.
- The midpoint is $(5, 3)$.
- Use point-gradient form: $y - 3 = (-3/4)(x - 5)$.
- An equivalent integer form is $3x + 4y = 27$.

FINAL ANSWER - 4 MARKS

$y - 3 = (-3/4)(x - 5)$, or $3x + 4y = 27$. Award one mark for each required idea and correct equation.

COMMON MISTAKE

Using point A or B. A bisector must pass through the midpoint.

Circles and tangents

Circle C: $x^2 + y^2 - 6x + 4y - 12 = 0$.

6(a) Complete both squares.

CHOOSE THE METHOD

Group x-terms and y-terms, complete each square, then move the constant.

- $x^2 - 6x = (x-3)^2 - 9$.
- $y^2 + 4y = (y+2)^2 - 4$.
- Substitute: $(x-3)^2 - 9 + (y+2)^2 - 4 - 12 = 0$.
- The constants total -25, so move them right.

FINAL ANSWER - 3 MARKS

$(x-3)^2 + (y+2)^2 = 25$. Award M1+M1+A1.

COMMON MISTAKE

Reading centre signs before completing the equation carefully.

6(b) State centre and radius.

CHOOSE THE METHOD

Compare with $(x-a)^2 + (y-b)^2 = r^2$.

- $x-3$ gives centre $x=3$.
- $y+2$ means $y-(-2)$, so centre $y=-2$.
- $r^2=25$, so $r=5$.

FINAL ANSWER - 2 MARKS

Centre (3,-2), radius 5. Award B1+B1.

COMMON MISTAKE

Giving centre (3,2). The sign inside the y bracket reverses.

6(c) Show P=(6,2) lies on C.

CHOOSE THE METHOD

Substitute P into the completed-square equation.

- $(6-3)^2 + (2+2)^2$.
- This is $3^2 + 4^2 = 9 + 16 = 25$.
- It equals the right side, so P satisfies the circle equation.

FINAL ANSWER - 2 MARKS

P lies on C. Award M1 for substitution and A1 for the demonstrated equality.

COMMON MISTAKE

Writing only 'yes'. The question says show, so the substitution must appear.

Radius, tangent and intersections

6(d) Find the radius gradient.

CHOOSE THE METHOD

Use centre (3,-2) and P=(6,2).

- $m = (2 - (-2)) / (6 - 3)$.

- This is $4/3$.

FINAL ANSWER - 2 MARKS

Radius gradient $4/3$. Award M1+A1.

COMMON MISTAKE

Using the radius length 5 as the gradient.

6(e) Find the tangent at P.

CHOOSE THE METHOD

The tangent is perpendicular to the radius and passes through P.

- Radius gradient = $4/3$, so tangent gradient = $-3/4$.

- Use P=(6,2): $y - 2 = (-3/4)(x - 6)$.

- Multiply by 4: $4y - 8 = -3x + 18$.

- Rearrange: $3x + 4y = 26$.

FINAL ANSWER - 3 MARKS

$y - 2 = (-3/4)(x - 6)$, or $3x + 4y = 26$. Award M1+M1+A1.

COMMON MISTAKE

Using the centre in the tangent equation. The tangent passes through P, not the centre.

6(f) Intersections with $y = -2$.

CHOOSE THE METHOD

Substitute the line into the circle equation.

- In $(x-3)^2 + (y+2)^2 = 25$, set $y = -2$.

- Then $(y+2)^2 = 0$, so $(x-3)^2 = 25$.

- $x - 3 = \pm 5$, giving $x = 8$ or $x = -2$.

FINAL ANSWER - 2 MARKS

The points are (8,-2) and (-2,-2). Award M1 for the equation and A1 for both points.

COMMON MISTAKE

Reporting only x-values. The question asks for points, so include $y = -2$.

Turn the checkpoint into progress

Complete this before moving to the next topic.

Q/part	My first wrong line	Error type	Correct rule

Topic	Score	Available	%
Quadratics		24	
Functions		10	
Coordinate geometry		12	
Circles		14	
Total		60	

RETEST RULE

For every topic below 80%, close this booklet and solve three new questions. For every topic below 60%, reread the relevant notes before the retest.

The three rules I must remember next time:
